BRONZE LAYER

Raw Data Ingestion & Source Preservation

SQL DATA WAREHOUSE PROJECT

SQL Server | Medallion Architecture

PURPOSE Document the layer workflow, key SQL patterns, validation checks, and implementation evidence.

1. Layer objective

The Bronze layer is the raw landing zone. It brings source-system files into SQL Server tables while keeping the incoming data as close to the original source as practical. This creates a traceable starting point for downstream transformations.

2. Source datasets

Source group	Expected CSV files
CRM	cust_info.csv • prd_info.csv • sales_details.csv
ERP	CUST_AZ12.csv • LOC_A101.csv • PX_CAT_G1V2.csv

Confirm exact filenames and capitalization against the source_crm and source_erp folders before publishing.

3. Ingestion workflow

Stage	Activity	Output
01	Prepare Bronze schema and destination tables	Empty Bronze tables
02	Load CSV files using BULK INSERT or the implemented loader	Raw records in SQL Server
03	Capture load counts and execution messages	Load evidence
04	Validate schema, row counts, and load errors	Verified raw layer

4. Example SQL pattern

Illustrative example only. Adjust the table, file path, column layout, and BULK INSERT options to match your executed script.

```
TRUNCATE TABLE bronze.crm_cust_info;

BULK INSERT bronze.crm_cust_info
FROM 'C:\path\to\source_crm\cust_info.csv'
WITH (
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '0x0a',
    TABLOCK
);
```

5. Validation checklist

- Confirm the SQL Server service account can access each source file.
- Check that destination column order and data types match the CSV structure.
- Compare source and loaded row counts.
- Review conversion errors, truncation, rejected rows, and unexpected NULLs.
- Confirm reload behavior is intentional before truncating existing data.

6. Implementation evidence

Evidence item	Complete with actual project value

Database and schema	[Database name] / bronze
Bronze tables	[List implemented table names]
Rows loaded	[Actual row count per table]
Execution duration	[Actual duration]
Execution date	[Date of successful run]

✅Completion target: all intended source files are loaded, the process can be rerun safely, and row-count/schema checks are recorded.

 Verification note: actual DDL, procedure names, BULK INSERT settings, and load results were not supplied. Replace examples and placeholders with the executed project details.